

STE-CLOTILDE, QC, Canada

WMO: 716140

Lat: 45.1667N

Lon: 73.6783W

Elev: 174

StdP: 14.60

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.7	-7.3	-20.6	1.8	-11.3	-15.5	2.4	-6.6	21.2	29.7	18.8	21.8	4.6	270	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	86.8	71.5	83.9	70.1	81.3	68.7	74.6	82.9	72.7	80.3	71.0	78.0	7.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.9	118.8	79.0	70.2	112.0	77.1	68.6	105.6	75.2	38.3	82.7	36.5	80.7	35.0	78.1	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	15.5	13.7	-21.9	91.5	7.4	2.5	-27.2	93.3	-31.5	94.7	-35.6	96.1	-41.0	97.9
			-22.2	77.7	7.2	1.9	-27.4	79.0	-31.7	80.1	-35.7	81.1	-41.0	82.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	44.3	15.7	18.8	28.7	42.8	55.8	64.7	69.1	67.1	59.7	47.5	36.0	24.0	(6)
(7)		DBStd	20.72	13.62	11.97	11.23	8.53	7.66	6.85	5.22	5.47	7.51	8.12	9.40	11.56	(7)
(8)		HDD50	4268	1063	875	663	247	28	0	0	0	8	147	429	808	(8)
(9)		HDD65	7941	1528	1295	1125	666	302	87	18	41	193	543	871	1272	(9)
(10)		CDD50	2183	0	0	3	32	206	443	591	531	298	71	7	1	(10)
(11)		CDD65	382	0	0	0	1	16	79	145	107	33	1	0	0	(11)
(12)		CDH74	3429	0	0	1	29	218	756	1215	878	315	17	0	0	(12)
(13)		CDH80	849	0	0	0	5	52	206	305	208	72	1	0	0	(13)
(14)	Wind	WSAvg	5.9	6.9	6.9	6.9	7.1	6.1	5.3	4.7	4.3	4.6	5.6	6.1	6.4	(14)
(15)	Precipitation	PrecAvg	37.8	2.4	2.1	2.3	3.1	3.3	3.7	3.9	3.7	3.5	3.7	3.0	3.1	(15)
(16)		PrecMax	44.0	4.4	3.9	4.0	5.6	5.7	7.0	5.7	6.9	7.0	7.6	6.6	6.0	(16)
(17)		PrecMin	32.2	1.0	0.8	0.7	0.5	1.1	1.5	2.1	1.1	1.2	1.3	0.8	1.1	(17)
(18)		PrecStd	3.9	0.9	0.7	0.8	1.3	1.4	1.4	1.1	1.3	1.3	1.7	1.1	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.8	49.8	63.6	78.1	85.9	89.6	90.5	89.4	87.5	76.3	64.6	53.3	(19)
(20)			MCWB	48.0	44.0	55.1	60.7	66.3	73.0	75.0	73.3	71.7	64.6	57.8	50.1	(20)
(21)		2%	DB	43.3	43.2	53.9	69.4	80.0	85.6	86.3	85.0	81.2	70.5	59.1	47.1	(21)
(22)			MCWB	40.3	39.1	46.4	55.4	63.9	70.0	72.5	71.3	69.3	60.9	54.1	43.9	(22)
(23)		5%	DB	39.0	39.3	47.8	63.2	75.2	81.9	83.3	81.9	77.5	65.4	54.2	42.6	(23)
(24)			MCWB	36.7	35.7	42.0	51.7	60.4	68.3	70.8	69.5	67.2	57.9	49.6	40.1	(24)
(25)		10%	DB	35.1	36.1	43.5	57.7	70.9	78.3	80.7	79.0	73.5	61.0	50.1	38.7	(25)
(26)			MCWB	33.1	33.7	38.8	48.0	59.1	66.2	69.2	68.4	64.8	55.5	46.1	36.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.8	44.3	56.3	63.9	70.2	75.8	77.4	75.7	74.1	67.4	59.2	50.5	(27)
(28)			MDCB	49.4	47.3	62.2	74.5	80.8	85.3	86.3	83.9	82.8	73.6	62.6	52.7	(28)
(29)		2%	WB	40.8	39.4	47.4	57.9	66.9	73.0	75.2	73.5	71.0	62.5	54.6	44.5	(29)
(30)			MDCB	42.2	42.6	52.4	67.1	75.3	81.7	83.2	81.8	78.1	68.2	58.6	46.5	(30)
(31)		5%	WB	36.7	36.4	42.7	52.9	63.9	70.7	73.0	71.6	68.7	59.0	50.4	40.3	(31)
(32)			MDCB	38.6	38.6	47.2	60.8	71.4	78.1	79.9	78.4	75.6	64.7	53.6	42.6	(32)
(33)		10%	WB	33.0	33.8	39.2	49.2	60.7	68.3	71.2	69.9	66.3	55.8	46.2	36.7	(33)
(34)			MDCB	34.9	36.0	43.3	56.2	68.5	75.4	77.9	76.4	72.3	59.9	49.8	38.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.2	17.7	17.0	18.7	20.9	19.8	19.4	20.2	21.1	17.6	14.8	14.2	(35)
(36)			MCDBR	17.8	18.7	22.9	29.5	28.5	24.8	23.0	23.8	25.4	24.6	21.5	16.5	(36)
(37)			MCWBR	16.7	15.9	16.5	17.6	14.6	12.6	11.9	12.3	14.7	16.0	17.1	14.6	(37)
(38)		5% WB	MCDBR	17.3	17.1	21.4	26.1	22.0	20.8	19.6	20.6	22.2	21.7	20.5	15.9	(38)
(39)			MCWBR	16.4	15.2	16.5	17.2	13.6	12.1	11.2	11.7	13.7	15.4	17.5	14.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.269	0.290	0.326	0.363	0.387	0.446	0.435	0.417	0.375	0.336	0.309	0.294	(40)	
(41)		taud	2.369	2.304	2.348	2.358	2.347	2.187	2.224	2.270	2.376	2.490	2.490	2.345	(41)	
(42)		Ebn at Noon	270	284	287	285	280	263	264	264	267	265	253	245	(42)	
(43)		Edh at Noon	23	30	34	37	38	45	43	40	33	26	21	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	455	745	1130	1382	1617	1764	1786	1586	1265	734	452	323	(44)	
(45)		RadStd	51	75	90	162	162	140	137	74	103	69	56	27	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+112	N/A

Nomenclature: See separate page